

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. University counseling combines relational curricula, cohort-specific fees, policy tables, and narrative regulations, for which a uniform evidence path may be unsuitable. We present CTU-Chat, a supervisor-routed multi-agent system that assigns queries to four specialists with scoped representations and tools: Neo4j for curricula, structured lookup and deterministic calculation for financial data, and hybrid document retrieval for regulations. A 1,960-decision semantic-neighbor stress test compares full-registry and Top- k single-agent baselines with routed generic and specialist gates across registries of 11–51 tools. At 51 tools, the specialist exceeds the global Top-10 baseline in decision-level end-to-end pass rate by 9.3 percentage points (95% CI [+2.0, +17.3]), but uses 70.8% more input tokens and 2.03 $\times$ the latency. Relative to the full-registry baseline, it uses 52.4% fewer input tokens, while its 4.7-point end-to-end difference has a CI containing zero. Positive high-minus-low interactions for selection and end-to-end pass rate also have CIs containing zero, so a scaling-rate advantage is not established. On a redesigned 100-query held-out set, the full retrieval stack achieves 0.7400 Hit@1, 0.8608 Recall@5, and 0.8333 MRR@10; the full generation configuration reaches 0.6534 Context Recall, 0.8636 Faithfulness, and 0.8325 Source Recall. These findings suggest that domain-specific disambiguation and heterogeneous evidence allocation can support institutional counseling under the evaluated conditions, with configuration-dependent accuracy and cost trade-offs.

Keywords: Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Can Tho University (CTU) serves nearly 49,000 undergraduate students in the 2025–2026 academic year [1]. Its scale and evolving regulations create advisory needs across academic and administrative areas. CAAS supports chat- and voice-based admission

counseling for information-technology programs [10], while REBot addresses academic-regulation advising through dense retrieval and category-guided graph reasoning [9]. Their reported evaluations support task-specific retrieval and processing, but each centers on an individual advisory domain, motivating a coordinated multi-domain architecture.

University counseling also spans heterogeneous knowledge forms: (1) **academic curricula** encode hierarchies and prerequisites; (2) **tuition schedules** vary by cohort and field; (3) **exemption policies** and (4) **scholarship policies** combine eligibility conditions with rates and thresholds; and (5) **administrative regulations** comprise lengthy narrative documents. A single representation or retrieval method may therefore be unsuitable across all domains, motivating representation-specific evidence acquisition.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [8,5,3], while agentic architectures support tool use and multi-agent coordination [20,18,16]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [14] further motivate deterministic calculation for rule-based financial queries.

Together with the representative literature synthesized in Section 2, these challenges motivate three gaps examined here: (1) limited evidence on bounded specialization under tool ambiguity; (2) limitations of uniform representations for heterogeneous knowledge; and (3) limited evidence on routing across isolated, representation-specific evidence paths.

The objective of this study is to develop and evaluate CTU-Chat for heterogeneous university counseling. To address the identified gaps, we formulate three research questions:

- **RQ1 (Domain-Specialized Tool Disambiguation under Registry Expansion):** How do full-registry and Top- k single-agent baselines compare with supervisor-routed generic and specialist gates in tool-selection reliability, decision-level end-to-end pass rate, and computational cost as semantically overlapping within-domain tools expand the registry from 11 to 51 tools?
- **RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations):** What differences in context and answer quality are observed between the full heterogeneous configuration, uniform retrieval baselines, and its component ablations?
- **RQ3 (Representation-Aware Retrieval Effectiveness):** How does the complete representation-aware evidence configuration compare with sparse, dense, hybrid, and reranked baselines in retrieval effectiveness, both overall and across query-complexity tiers?

We hypothesize that domain-scoped candidate retrieval and contrastive specialist policies can reduce confusion among semantically adjacent tools relative to a global Top- k single-agent gate. This hypothesis is limited to the evaluated registry expansions and does not predict multi-agent dominance over a full-registry monolithic baseline on every accuracy or efficiency measure. CTU-Chat coordinates four specialists for academic, financial, scholarship, and general administrative inquiries, combining graph retrieval, structured access, deterministic calculation, and hybrid document retrieval. We evaluate it through retrieval stacking, end-to-end generation and ablation, and controlled tool-ambiguity stress tests against matched single-agent baselines.

Specifically, this work provides three primary contributions:

- **C1 (Domain-Specialized Multi-Agent Architecture):** We implement a centralized supervisor coordinating four bounded specialists with isolated evidence representations, deterministic route repair, and partitioned tool privileges, and evaluate registries of up to 51 tools.
- **C2 (Heterogeneous Knowledge Allocation):** We model relational curricula as property graphs, bind statutory fee and exemption lookup to deterministic arithmetic, and index narrative regulations through parent-resolved document retrieval.
- **C3 (Tool-Ambiguity Stress Test and End-to-End Evaluation):** We evaluate semantic-neighbor ambiguity across 1,960 decisions, retrieval stacking across five configurations, and generation quality across seven configurations (2, 100 turns) on a redesigned 100-query held-out benchmark.

The remainder of this paper is organized as follows. **Section 2** reviews related work and summarizes the research gaps. **Section 3** presents the CTU-Chat architecture, evidence-acquisition methods, and domain-specialized agents. **Section 4** reports the experimental setup, results, and discussion. Finally, **Section 5** concludes the paper and outlines future work.

2 Related Work

2.1 Vietnamese NLP Foundations and Linguistic Specifics

Unlike whitespace-delimited languages, Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11]; 85% of Vietnamese word types contain at least two syllables. While domain-adapted bi-encoders like PhoBERT capture semantic paraphrasing, Vietnamese higher education decrees rely heavily on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes. This dual characteristic motivates coupling syllable-aware dense embeddings with exact lexical token matching to mitigate both semantic drift and alphanumeric recall failures.

2.2 Sparse and Dense Retrieval

Lexical Retrieval (BM25): Sparse keyword search using BM25 [15] scores documents via term frequency (tf) and inverse document frequency (IDF) over an inverted index ($k_1 = 1.5, b = 0.75$). BM25 provides reliable exact token matching for rigid alphanumeric identifiers (decision codes, cohort labels, course codes), but cannot capture semantic paraphrases.

Dense Semantic Retrieval: Dense retrieval systems [7,8] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

2.3 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position: $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$ ($k = 60$), while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [6]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.4 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples (h, r, t) [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious or hallucinated relationships, especially when narrative documents lack explicit relational structure. Therefore, this work adopts selective rather than universal graph construction.

2.5 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,16]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

2.6 Educational and University Counseling Systems

Educational chatbots assist in advising and learning support [17], with recent systems like REBot [9] combining text retrieval with category-guided GraphRAG for academic regulations. CTU-CHAT departs from prior advisory systems across three key aspects: (1) *Task Scope*: while representative prior advisory systems often emphasize single-domain text-QA or focus primarily on narrative regulations, CTU-CHAT models 113 curricula as relational property graphs and incorporates deterministic arithmetic engines (`tin_h_toan_hoc_phi`, `tin_h_tien_hoc_bong`) for cohort-dependent tuition, exemptions, and scholarship thresholds; (2) *Multi-Agent Routing*: rather than a monolithic classifier-guided pipeline, a LangGraph supervisor dynamically routes queries to four bounded specialists with isolated evidence spaces and asymmetric tool privileges; and (3) *Metric Incommensurability*: prior benchmarks evaluate traditional classification accuracy and binary Q&A F_1 -scores, precluding direct numerical comparison with CTU-CHAT, which evaluates end-to-end generative fidelity via Ragas and bounded multi-agent tool execution benchmarks.

2.7 Comparative Synthesis and Research Gaps

Synthesizing established retrieval and reasoning paradigms highlights operational trade-offs for university counseling [2,3,6,7,15,16,18,20]. While individual techniques address specific needs—lexical matching for exact regulatory tokens, dense embeddings for semantic paraphrasing, and GraphRAG for relational prerequisites—the representative approaches reviewed here do not jointly address the three requirements of multi-domain institutional advising.

This analysis identifies three operational gaps in automated university counseling: (1) *Limited evidence on bounded specialization under tool ambiguity*: monolithic pipelines and unconstrained agent meshes may lack explicit execution guardrails, while it remains unclear when domain-specific policies improve upon global single-agent selection as semantically adjacent tools accumulate; (2) *Limitations of uniform representation*: placing relational curricula, tabular tuition matrices, and narrative legal text in a single retrieval space may fragment prerequisite evidence or introduce semantic drift; and (3) *Absence of representation-aware routing*: the representative systems reviewed here do not directly address dynamic steering across isolated representation pathways while offloading statutory arithmetic to deterministic engines. These requirements motivate the centralized supervisor-routed architecture and comparative tool-disambiguation evaluation detailed in Sections 3–4.

3 Proposed Model

This section presents the architecture, algorithms, and evidence-acquisition mechanisms of CTU-CHAT. The proposed model is organized around three design principles: *centralized agent coordination*, in which a contract-checked supervisor dispatches each query to one domain agent without peer-to-peer delegation; *heterogeneous knowledge allocation*, which assigns each information structure to a corresponding evidence representation; and *representation-aware evidence routing*, which selects either a direct graph-tool path or an intent-scoped retrieval path according to the assigned domain. Each *specialist* is configured with domain-specific prompts, a scoped evidence path, and a restricted tool registry.

3.1 Overall Proposed Architecture

Figure 1 illustrates the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

Application-Level Specialization of LangGraph StateGraph: CTU-CHAT specializes LangGraph through typed state, an intent-owner contract, isolated paths, and partitioned tool registries. The contract checks structured supervisor output before dispatch; specialists then operate without peer delegation and see only their assigned tools.

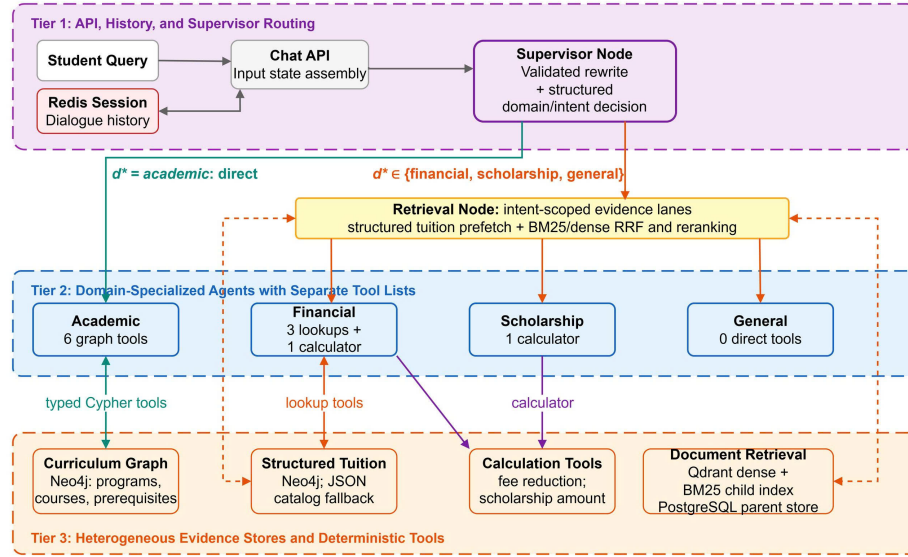


Fig. 1. Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked and may be repaired before conditional dispatch (repair is internal, not a separate node). Academic queries enter the graph-enabled agent, whereas other domains pass through intent-scoped retrieval. Solid and dashed links denote direct tool access and retrieval-node evidence acquisition, respectively.

- **Node-scoped state:** AgentState retains the validated rewrite, raw and repaired routes, repair reason, evidence, and response; each node writes only its corresponding fields.
- **Conditional routing:** Deterministic edges send academic queries directly to the graph agent and all other queries through retrieval before dispatch to the selected financial, scholarship, or general agent.
- **Acyclic orchestration:** Specialists have no peer edges and return to END; their internal ReAct tool interactions are separate from the outer graph.
- **Asymmetric tools:** Academic, financial, and scholarship agents receive scoped lists, while the general agent has none. One bounded retry names an omitted required tool; invalid or missing parameters trigger clarification.

3.2 Neo4j Knowledge Graph Specification and Cypher Tool Mechanics

Schema defined by the ingestion layer: The Neo4j graph contains over 1,500 nodes and 3,800 edges across 113 undergraduate curricula and fee schedules. Its 10 node labels cover curriculum entities (e.g., Program, Course, and Block) and tuition entities (TuitionFee, TuitionPolicy, and ExemptionBasisRate); 13 typed relationships

encode containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases. Narrative regulations remain in the document store.

Cypher tool mechanics: The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies the tool arguments. The tool normalizes identifiers or names and calls a fixed, parameterized Cypher query in the graph service. The tools implement three query classes:

- *Attribute lookup and search:* `tra_cuu_nganh` and `tim_nganh` match program codes or normalized program, unit, and course names; `so_sanh_nganh` applies the same lookup before comparing two programs.
- *Structural traversal:* `xem_chuoi_tien_quyet` follows `REQUIRES` paths up to ten hops and returns direct `PARALLEL_WITH` courses. `mon_chung_giua_nganh` computes a course intersection across two program–block paths, and `tim_nganh_co_mon` traverses those paths in reverse.
- *Structured financial lookup:* parameterized queries match `TuitionFee`, `TuitionPolicy`, and `ExemptionBasisRate` properties and their program relationships. This graph lookup is separate from the BM25–dense hybrid document retrieval path used for narrative regulations.

This design binds agent actions directly to deterministic graph operations: the LLM only selects the appropriate tool and supplies its arguments, while the query topology remains fixed and parameterized within the graph service. The multi-agent workflow therefore incorporates graph evidence through bounded operations, reducing exposure to the syntactic and semantic errors associated with unconstrained Cypher generation.

3.3 Algorithmic Foundations: Routing and Deterministic Arithmetic

For a query q and its most recent dialogue predecessor, the supervisor first attempts a validated standalone rewrite. It then predicts domain d_{raw} and intent ι_{raw} under a structured schema, falling back to `general/other` on parsing failure. The ownership contract repairs strong rule conflicts or incompatible domain–intent pairs and records both raw and repaired decisions in the shared state. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback. The selected specialist receives the resulting state and terminates at `END`.

Deterministic Tuition Reduction Arithmetic: For fee-reduction requests, CTU-CHAT augments the LLM with the external deterministic tool `tin_hoc_phi`. The LLM interprets the request and supplies three parameters—the actual per-credit tuition T , the per-credit exemption basis B , and the percentage entitlement p —but does not perform the arithmetic through free-form text generation. The external tool validates the percentage range and applies the fixed calculation

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (1)$$

where D is the reduction per credit and P is the remaining amount per credit. Monetary inputs are expected in VND per credit. The tool enforces $0 \leq p \leq 100$ and clamps P at zero; values outside the percentage range return an error.

Rather than a new general-purpose algorithm, the contribution lies in composing validated rewriting, constrained routing, representation-specific dispatch, and deterministic arithmetic within the multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

This layer implements *heterogeneous knowledge allocation*: rather than forcing all institutional knowledge into a uniform document index, CTU-CHAT preserves relational curriculum knowledge in Neo4j, accesses cohort-dependent tuition data through structured lookup, and retains narrative regulations and scholarship policies in document retrieval.

The access policy maps program attributes to Neo4j property lookup, prerequisite and shared-course questions to graph traversal, regulation identifiers to BM25, paraphrased regulations to hybrid RRF with reranking, and fee or scholarship arithmetic to deterministic Python tools.

At runtime, academic queries enter the graph agent directly; other domains enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback. The repaired intent then selects metadata-filtered document lanes, and packed evidence is passed only to the selected specialist.

Hybrid Document Retrieval Path. For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; their parent sections are loaded from PostgreSQL and fused using $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} (60 + r_s(d))^{-1}$. The full configuration pools intent-governed candidates with the top 20 unfiltered hybrid candidates and top BM25 anchors, deduplicates them, and reranks with BAAI/bge-reranker-v2-m3. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is then added under fixed quotas and up to seven passages are packed for generation.

Parent sections use a 2,800-character target with 100-character overlap; child chunks use a 400-character target with 50-character overlap.

4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT.

4.1 Experimental Benchmark and Protocols

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded semi-automated protocol over official Can Tho University (CTU) statutes. Candidate queries with diverse conversational forms were generated from statutory articles and reviewed for ground-truth intents, passages, required facts, and answers. The 100-query development split supports index calibration, prompt engineering, and failure analysis. The separately constructed 100-query held-out set follows a predefined complexity distribution of 40 direct single-hop, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries. It includes 51 formal and 49 colloquial formulations; 56 queries have one gold source and 44 require multiple gold sources.

An audit using entity inventories separate from the development split found no exact cross-split duplicates and a maximum 5-gram Jaccard similarity of 0.1875. Gold sources and required facts were reviewed against 244 institutional documents and 559 structured tuition records. The held-out set contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains and are analyzed separately. Tables 1 and 3 report results on this held-out set.

Scenario 2 used Ragas 0.4.3 with Answer Relevancy strictness 1. All 10,500 required Ragas values were valid. For uncertainty estimates, the three repetitions were first averaged within each query; 95% confidence intervals then used 10,000 query-level bootstrap resamples over the 100 independent queries.

Evaluated Model and Infrastructure: The evaluation used Gemini 2.5 Flash Lite through Google Cloud Vertex AI at temperature 0.0. Scenario 2 evaluated each query in three repetitions, producing 2,100 answers (7 configurations $\times$ 100 queries $\times$ 3). Neural reranking used an NVIDIA GeForce GTX 1650 GPU (4 GB VRAM). Neo4j 5.20 with APOC stored relational curricula; PostgreSQL stored parent chunks (2,800 characters, 100 overlap); and Qdrant indexed child chunks (400 characters, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. The system used bge-reranker-v2-m3, LangGraph StateGraph, and Redis 7.

Scenario 3 Semantic-Neighbor Protocol: The tool-ambiguity experiment fixes 50 production-tool cases (30 academic and 20 financial) while expanding an 11-tool core with 40 evaluation-only, operationally plausible semantic neighbors distributed across ten production tool families. Adding zero, two, or four neighbors per family produces registries of 11, 31, and 51 tools. Four configurations are compared: a monolithic full-registry gate, a global Top-10 single-agent gate, a supervisor-routed Top-10 gate with a generic selection policy, and a supervisor-routed Top-10 specialist gate with contrastive use/avoid rules derived from the tool contracts. Each primary case is evaluated under three counterbalanced tool-order blocks, yielding 1,800 decisions. A separate 40-case coverage suite makes every semantic-neighbor tool the gold target once at the 51-tool level, adding 160 decisions and producing 1,960 total records. To isolate downstream selection, 190 live supervisor decisions are shared across routed configurations and ambiguity levels for the same case and order block. Paired differences and high-minus-low interactions use 10,000 case-level bootstrap resamples after averaging repetitions within each case. The run contains 19 runtime-error records and four schema-guard failures; these records are retained as failed outcomes rather than excluded from the reported rates.

4.2 Evaluation Metrics

Evaluation Metrics: Scenario 1 reports source-document-level Hit@ k ($k \in \{1, 3\}$), Precision@5, Recall@5, and MRR@10. In Scenario 3, agent and intent accuracy require exact labels, and Macro-F1 is unweighted across routing classes. Selection requires the gold tool; Argument EM accepts only contract-equivalent arguments; Result Accuracy requires the normalized tool output to match its deterministic oracle. Decision-level E2E Pass is the conjunction of selection, arguments, result, and bounded execution, where bounded means at most one tool call with no timeout or runtime error. For adversarial cases, Tool Gate requires suppressing the tool and Safe Result requires a predefined clarification or refusal phrase. Gold@10 is gold-tool shortlist recall; Visible is the mean

exposed-tool count. Paired CIs and high-minus-low interactions use 10,000 case-level bootstrap resamples after averaging repetitions within each case.

Scenario 2 evaluates Top-7 synthesis using Ragas [4]: Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), Answer Correctness (AC), and Faithfulness (Faith). Required-Fact Coverage (Fact Cov.) is the fraction of author-reviewed required facts found after normalized string and numeric matching. Source Recall measures gold-source coverage, while Source Average Precision (Source AP) evaluates gold-source ranking. Gold references, temperature-zero decoding, and programmatic Fact Cov. reduce reliance on the LLM judge [21,4].

4.3 Experimental Scenarios and Benchmark Results

We evaluate CTU-CHAT through retrieval component stacking (RQ3), end-to-end generation and ablation (RQ2), and multi-agent coordination and tool reliability (RQ1).

Scenario 1 — Retrieval Component Stacking (E1–E5): Scenario 1 compares five cumulative retrieval configurations, from sparse and dense baselines to hybrid fusion, neural reranking, and the proposed intent-guided evidence stack, at aggregate and complexity-stratified levels on the 100-query held-out set.

Table 1 shows that E4 and E5 exceed the sparse, dense, and hybrid baselines on every aggregate metric. E4 reaches 0.7000 Hit@1 and 0.8054 MRR@10. The full intent-governed configuration E5 records the highest values: 0.7400 Hit@1, 0.9200 Hit@3, 0.3020 Precision@5, 0.8608 Recall@5, and 0.8333 MRR@10. Its higher latency (21.86 s vs. 15.37 s for E4) is consistent with additional intent-governed retrieval and graph-processing stages.

Table 1. Scenario 1: Progressive retrieval component stacking on the redesigned Held-Out Test Set ($N = 100$).

Retrieval Configuration	Hit@1	Hit@3	P@5	Recall@5	MRR@10	Latency (ms)
E1: BM25 Lexical Baseline	0.5300	0.7700	0.2060	0.6167	0.6695	7.08
E2: Dense Vector Semantic	0.4600	0.6500	0.1760	0.5242	0.5604	33.66
E3: Hybrid RRF (BM25 + Dense)	0.5200	0.7900	0.2240	0.6625	0.6725	37.57
E4: Hybrid + Neural Reranker	0.7000	0.9100	0.2540	0.7508	0.8054	15,366.33
E5: CTU-Chat Full Retrieval	0.7400	0.9200	0.3020	0.8608	0.8333	21,857.56

E5 matches or exceeds E4 on both metrics in every complexity tier. The largest Recall@5 differences occur for entity-comparison (28.33 percentage points; 0.9500 vs. 0.6667) and cross-domain queries (13.33 percentage points; 0.8250 vs. 0.6917). For the five temporal queries, E5 obtains 1.0000 on both metrics.

Scenario 2 — Controlled Retrieval-Conditioned Generation and Modular Ablation (T1–T7): Scenario 2 compares T4 with three retrieval baselines (T1–T3) and three

Table 2. E4–E5 retrieval results by held-out query complexity.

Complexity Tier	N	Hit@1		Recall@5	
		E4	E5	E4	E5
Direct single-hop	40	0.6500	0.6750	0.8500	0.9250
Within-domain multi-hop	20	0.7000	0.7000	0.7000	0.7583
Cross-domain	20	0.7500	0.8000	0.6917	0.8250
Entity comparison	10	0.8000	0.9000	0.6667	0.9500
Temporal	5	0.8000	1.0000	0.9000	1.0000
Adversarial	5	0.6000	0.6000	0.4167	0.5833

Table 3. Scenario 2: Controlled retrieval-conditioned generation and modular ablation on the redesigned Held-Out Test Set ($N = 100$, $R = 3$, 2,100 total evaluation turns, Top-7 context).

Cfg.	CP	CR	AR	AC	Faith	Fact Cov.	Src. R	Src. AP
T1	0.2522	0.5349	0.3901	0.4615	0.8042	0.5924	0.6358	0.4637
T2	0.1626	0.2648	0.2370	0.3372	0.5817	0.5179	0.5342	0.4094
T3	0.2071	0.4693	0.3972	0.4361	0.7486	0.5792	0.6892	0.5015
T4	0.3511	0.6534	0.4468	0.4785	0.8636	0.6331	0.8325	0.7316
T5	0.2248	0.5192	0.3425	0.4330	0.7618	0.6056	0.7892	0.6834
T6	0.3765	0.5791	0.4719	0.4913	0.8376	0.6184	0.6925	0.5939
T7	0.3541	0.6299	0.4304	0.4785	0.8427	0.6345	0.8208	0.7277

Configurations: T1: BM25; T2: dense; T3: hybrid RRF; T4: full governed multi-evidence configuration; T5: without cross-reranker; T6: without Neo4j grounding; T7: without subspace governance. *Metrics:* Fact Cov.: Required-Fact Coverage; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Faith: Faithfulness; Src. R: Source Recall; Src. AP: Source Average Precision. Means use 300 query–repetition observations; CIs average repetitions per query and bootstrap 100 queries. All metrics are in $[0, 1]$; higher is better.

ablations (T5–T7) over 2,100 turns. Table 3 shows that T4 exceeds BM25 on all eight metrics, including Faithfulness (0.8636 vs. 0.8042) and Fact Cov. (0.6331 vs. 0.5924). T4 leads Context Recall, Faithfulness, Source Recall, and Source AP; T6 leads Context Precision, Answer Relevancy, and Answer Correctness, while T7 has slightly higher Fact Cov. (0.6345).

Scenario 3 — Multi-Agent Routing, Specialist Tool Reliability, and Semantic Tool Ambiguity: Scenario 3 evaluates coordination, bounded safety, and selection among semantically adjacent tools (Table 4). Panel A scores raw supervisor output before route repair; Panels B–C test isolated specialist gates and adversarial inputs. Panel D reports the 1,800-decision fixed-case semantic-neighbor experiment across 11, 31, and 51 tools; the additional 160-decision coverage suite is reported in the accompanying analysis.

Table 4. Scenario 3: Routing, specialist tool reliability, adversarial robustness, and semantic-neighbor tool ambiguity under a maximum of one tool call per decision ($\text{max_tool_calls} = 1$).

Panel A: Raw Supervisor Routing and Specialist Selection ($N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$)							
Router Variant	Runs (N)	Agent Acc.	Macro-F1	Intent Acc.	Bounded	Routing Paradigm	
Rule-Based Router	100	85.0%	69.2%	69.0%	100.0%	Heuristic Regex	
Raw LLM Supervisor (CTU-CHAT)	300	97.0%	97.8%	95.0%	100.0%	Structured Pydantic	
Panel B: Specialist Tool Execution Reliability ($N = 90$ runs across 30 test cases, 3 repeated runs)							
Specialist Tool Function	Runs (N)	Selection	Arg. EM	Result Acc.	E2E Pass	Bounded	Paradigm
tuition_lookup	30	100.0%	100.0%	100.0%	100.0%	100.0%	Structured Catalog
scholarship_calculation	30	86.7%	76.7%	86.7%	76.7%	100.0%	LLM Tool Selection
tuition_reduction_calculation	30	100.0%	100.0%	100.0%	100.0%	100.0%	Arithmetic Tool
Panel C: Robustness and Bounded Failure Handling ($N = 60$ runs under malformed/adversarial inputs)							
Adversarial Gate	Runs (N)	Agent Acc.	Tool Gate	Safe Result	E2E Pass	Bounded	Notes
Failure Gate Robustness	60	100.0%	100.0%	98.3%	98.3%	100.0%	–
Panel D: Semantic-Neighbor Tool Ambiguity ($N = 50$ fixed cases, 3 reps, $N = 1,800$)							
Registry	Configuration	Visible	Gold@10	Selection	E2E Pass	Input Tok.	Latency (ms)
11 tools	Full-registry single	11.0	–	94.0%	86.0%	1,663	1,098
	Global Top-10 single	10.0	100.0%	93.3%	83.3%	1,541	1,148
	Routed generic	5.1	98.0%	90.7%	80.0%	1,204	1,889
	Routed specialist	5.1	98.0%	93.3%	85.3%	1,441	1,983
31 tools	Full-registry single	31.0	–	91.3%	84.7%	2,706	1,157
	Global Top-10 single	10.0	98.0%	90.0%	81.3%	1,149	1,033
	Routed generic	9.8	98.0%	88.0%	79.3%	1,396	1,958
	Routed specialist	9.8	98.0%	93.3%	88.7%	1,830	2,015
51 tools	Full-registry single	51.0	–	92.7%	83.3%	3,722	1,289
	Global Top-10 single	10.0	96.0%	86.7%	78.7%	1,036	997
	Routed generic	9.8	98.0%	91.3%	82.7%	1,340	1,907
	Routed specialist	9.8	98.0%	92.7%	88.0%	1,770	2,021

Note: Panels A–C evaluate raw routing, isolated tool gates, and adversarial robustness. Panel D uses 50 fixed production-tool cases across 11, 31, and 51 tools with a distilled supervisor prompt. Gold@10 is gold-tool shortlist recall at $k = 10$; Visible is the mean active tool count. Bold marks the highest Selection/E2E value and the lowest input-token/latency value within each registry. Paired intervals use 10,000 case-level bootstrap resamples. The separate coverage suite makes each of 40 semantic-neighbor tools the gold target once.

4.4 Discussion

Answering RQ1: Before production route repair, the evaluated raw structured supervisor achieves 97.0% routing accuracy, 97.8% Macro-F1, and 95.0% intent accuracy, compared with 85.0%, 69.2%, and 69.0% for rule-based routing. The isolated deterministic tuition tools achieve 100.0% decision-level end-to-end pass rates, whereas scholarship calculation reaches 76.7% with 86.7% tool selection; Panels A–C remain bounded in all tested runs. Panel D evaluates behavior under increasing intra-domain semantic-neighbor density. The routed specialist is the only evaluated configuration whose decision-level end-to-end pass rate improves between the 11- and 51-tool endpoints (85.3% $\rightarrow$ 88.0%, $\Delta = +2.7$ pp), whereas both single-agent configurations decline: full-registry single 86.0% $\rightarrow$ 83.3% ($\Delta = -2.7$ pp) and global Top-10 single 83.3% $\rightarrow$ 78.7% ($\Delta = -4.6$ pp). At the 51-tool registry, the specialist has a +6.0-pp selection difference from the global Top-10 single-agent baseline (95% CI $[-1.3, +14.0]$) and a +9.3-pp decision-level end-to-end difference (CI $[+2.0, +17.3]$). Relative to the routed generic gate, the specialist has higher E2E point estimates at all three registry sizes; the difference is +5.3 pp (CI $[+0.7, +10.7]$) at 51 tools, consistent with a possible contribution from the contrastive domain policy. Selection accuracy changes from 93.3% to 92.7% for the specialist ($\Delta = -0.6$ pp), while the Top-10 single agent declines by 6.6 pp between the same endpoints.

At 51 tools, the specialist uses 1,770 input tokens, including 445 routing tokens. This is 52.4% fewer than full-registry single (3,722), but 70.8% more than Top-10 single (1,036); its latency is also $2.03\times$ Top-10 (2,021 vs. 997 ms). From 11 to 51 tools, specialist token cost grows $1.23\times$ versus $2.24\times$ for full-registry single, but three registry sizes do not establish an asymptotic rate. Specialist-vs.-Top-10 high-minus-low interactions are +6.0 pp for selection (CI $[0.0, +13.3]$) and +7.3 pp for E2E (CI $[0.0, +15.3]$); because both intervals touch zero, differential degradation remains inconclusive. In the one-repetition coverage suite, specialist and full-registry selection are both 92.5%, with E2E point estimates of 82.5% and 80.0%. Thus Panel D supports a fixed-level E2E advantage over Top-10 at 51 tools, not a conclusive scaling-rate advantage.

Answering RQ2: T4 exceeds T1–T3 and the no-reranker T5 on all eight metrics, although these contrasts do not isolate individual mechanisms. Its Faithfulness difference from T5 is 0.1018 (query-level 95% bootstrap CI $[+0.0294, +0.1762]$). Comparisons with other ablations are mixed: relative to T6, T4 is higher on Context Recall, Faithfulness, Source Recall, Source AP, and Fact Cov., but lower on Context Precision, Answer Relevancy, and Answer Correctness. Relative to T7, T4 is higher on five metrics, T7 is slightly higher on Context Precision and Fact Cov., and Answer Correctness ties at reported precision; all five paired Ragas CIs include zero. These results support component-specific associations rather than uniform causal effects.

Answering RQ3: E4, which adds neural reranking to hybrid retrieval, exceeds E3 on every aggregate metric. E5 records the highest aggregate values and matches or exceeds E4 on Hit@1 and Recall@5 in every complexity tier. Because E5 jointly adds intent-governed lanes, graph evidence, and quota-based packing, these differences cannot be attributed to an individual component; they are nevertheless consistent with improved multi-source evidence aggregation in the evaluated setting.

4.5 Threats to Validity and Limitations

The evaluation is limited to CTU and 100 held-out queries, with five temporal and five adversarial cases; Gemini 2.5 Flash Lite may also introduce generator–judge bias. Scenario 3 uses 50 fixed cases and evaluation-only semantic-neighbor contracts. Its coverage suite has one repetition, and shared supervisor decisions control downstream comparisons but omit independent route variability. The run contains 19 runtime-error records and four schema-guard failures, all retained as failed outcomes; runtime errors occur in the routed configurations and therefore remain a potential implementation-level confound. While the high-minus-low interaction estimates are positive, their confidence intervals include zero, so the differential degradation rate warrants replication with larger samples. Across the tested registry sizes, the routed specialist has approximately 1.6–2.0× the decision latency of the single-agent configurations because it uses a two-hop LLM sequence (supervisor then gate), despite a lower token count than the full-registry monolithic agent at 51 tools. One-decision gates do not reproduce the complete repair/retry path. Live fault injection, naturally evolving registries, and multi-institutional validation remain future work.

5 Conclusion and Future Work

This study developed and evaluated CTU-CHAT, a supervisor-routed multi-agent RAG framework for heterogeneous university counseling. Three findings answer the research questions. For RQ1, at 51 tools the routed specialist exceeds the global Top-10 single-agent baseline by 9.3 percentage points in decision-level end-to-end pass rate (95% CI [+2.0, +17.3]), but uses 70.8% more input tokens and 2.03× the latency. Relative to full-registry single-agent execution, it uses 52.4% fewer input tokens, while its 4.7-point end-to-end difference has a CI containing zero. The evidence therefore supports a fixed-registry advantage over Top-10 at 51 tools, not an asymptotic scaling-rate advantage. For RQ2, the full heterogeneous configuration exceeds the three retrieval baselines and the no-reranker ablation on all eight metrics, including a 0.1018 Faithfulness difference from the latter (95% CI [+0.0294, +0.1762]); comparisons with the remaining ablations are mixed and support component-specific associations rather than uniform effects. For RQ3, the complete retrieval configuration records the highest aggregate scores and matches or exceeds the reranked baseline across all complexity tiers, although its bundled mechanisms prevent attribution to any single component.

Taken together, these findings are consistent with the interpretation that representation-specific evidence paths—graph traversal for curricula, deterministic calculation for statutory fees, and hybrid retrieval for narrative regulations—can accommodate heterogeneous institutional knowledge under the evaluated conditions. The study contributes an implemented architecture and an evaluation protocol that expose both reliability gains and operational costs. In deployment, specialist routing is supported when tool ambiguity and bounded evidence access are priorities, whereas a Top- k single-agent gate remains preferable when latency and input-token cost dominate. These implications remain limited by the offline, single-institution benchmark, automated evaluation, implementation-level failures, and dependence on upstream classification and route repair.

Future work should (1) reduce routing latency through parallel candidate dispatch and prompt compression; (2) test multi-institutional and bilingual benchmarks with naturally evolving tool registries; and (3) conduct human-in-the-loop field trials of student trust, advisory efficacy, and practical usability.

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